a

Single-cell NOMe-HiC contact

accessible / methylated

closed / unmethylated

chromatin contact

locus *i*

locus *j*

k_i/n_i

k_j/n_j

GCH (accessibility) · HCG (CpG methylation)

b

Per-end state posterior

p_j

p_i

density

P(open)

$p = P(\text{open} \mid k, n)$

Beta–Binomial mixture

c

Soft pairwise co-state

co-open

$p_i\, p_j$

co-closed

$(1-p_i)(1-p_j)$

mixed

$p_i(1-p_j) + (1-p_i)p_j$

soft expected counts (no thresholding)

d

Long-range vs local baseline

aggregate within 250-kb bin

d_0 (≈20 kb)

contact distance

unlinked
($< d_0$, baseline)

linked
(long-range)

e

Signed long-range concordance

$$\Delta_{\text{signed}} = (\hat{\pi}_{\text{co-open}} - \hat{\pi}_{\text{co-closed}})_{\text{linked}}$$

$$- (\hat{\pi}_{\text{co-open}} - \hat{\pi}_{\text{co-closed}})_{\text{unlinked}}$$

+

−

cells

bins

f

Cell-graph denoising → cell types

RWR over
cell *k*-NN graph

UMAP

type A

type B